Indian Institute of Technology Roorkee

DAL-311 — Network Science

Trade-Network Resilience and the Strait of Hormuz: A Multi-Layer Exposure Analysis of Five Commodity Networks (2020–2024)

Group members

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Abstract

The Strait of Hormuz carries roughly a fifth of the world's seaborne energy trade, and a closure or partial blockade would propagate asymmetrically across commodity-specific trade networks. We assemble six years (2020–2025) of bilateral monthly trade flows, tanker-capacity indicators and benchmark prices to build five directed weighted commodity networks — wheat, ammonia, urea, LPG propane, and LPG butane — and measure their exposure to a Hormuz disruption. Using community detection, weighted topological metrics and targeted-attack simulations, we show that LPG butane is the most fragile layer: removing Hormuz exporters wipes out 48.5 % of trade volume and 32.4 % of weighted global efficiency in 2024. A bivariate Vector AutoRegression and a Spatial Durbin panel model identify a small but consistent network- contagion coefficient ($\rho \approx 0.11$) and a statistically significant Granger link from Hormuz tanker capacity to ammonia prices. A scenario engine combining structural disruption with the spatial-econometric multipliers projects that under a six-month realignment scenario, India alone faces an annualised import-value loss of \$7.6 billion — driven entirely by its ≈ 97 % Hormuz dependence on LPG. We discuss the methodological limits of the econometric channel and the policy implications of LPG concentration.

1. Introduction and problem statement

The Strait of Hormuz is the maritime corridor between the Persian Gulf and the Gulf of Oman, through which an estimated 17–20 % of global oil-equivalent energy and over a quarter of global LPG trade transits (U.S. Energy Information Administration, 2023). Repeated geopolitical episodes (the 2019 tanker seizures, the 2024 Iran–Israel exchange, recurring threats to "close" the strait) have repeatedly raised the question: *how exposed is each commodity-specific trade network to a Hormuz disruption, and how would a shock propagate across countries?*

Research questions. This project tackles five concrete questions:

1. **Structure.** Are the five commodity networks topologically similar, or do they have distinct exposure signatures to Hormuz exporters?
2. **Exposure.** Which countries are most exposed, and is "exposure" best captured by direct bilateral share, by global centrality, or by an information-flow measure?
3. **Resilience.** How much trade volume and connectivity is lost when Hormuz exporter rows are removed from each network?

4. **Dynamics.** Does Hormuz tanker capacity Granger-cause commodity price movements, and does shock propagation across the network amplify imports' co-movement?
5. **Scenarios.** Under a 14-day shock, a 30–45 day rupture, and a 6-month realignment of trade routes, what is the projected country-level dollar impact on imports?

Why a multi-layer treatment. Treating "trade" as a single flow understates the very different role Hormuz plays for, e.g., LPG (where Gulf states are dominant exporters) versus wheat (where they are net importers). The five layers chosen here separate energy (LPG butane, LPG propane), agriculture (wheat) and fertilizer (ammonia, urea) so that the structural effects of a Hormuz disruption are not averaged out across heterogeneous flows.

2. Data sources and collection

The pipeline draws on five public, machine-readable databases:

- **UN Comtrade Plus** (United Nations, 2025): bilateral monthly trade flows by HS code. We extract HS 1001 (wheat), HS 2814 (ammonia), HS 3102 10 (urea), HS 2711 12 (LPG propane) and HS 2711 13 (LPG butane) for 2020–2025 (204 019 reporter-partner-month observations).
- **IMF PortWatch** (International Monetary Fund, 2025): daily counts and deadweight tonnage (DWT) of vessels transiting global chokepoints. We use the Hormuz daily tanker series, aggregated to monthly n_{tanker} and capacity DWT.
- **World Bank Pink Sheet** (World Bank, 2025): global benchmark prices — US HRW wheat, urea, DAP (used as ammonia proxy), and LNG Japan (used as butane proxy). Monthly, 2020-01 to 2026-02.
- **FRED** (Federal Reserve Bank of St. Louis, 2025): daily Mont Belvieu propane spot, averaged to monthly. Pre-2021-03 values are backfilled from a scaled LNG Japan index (overlap-mean calibration on 2021-03..2021-12).
- **JODI World Database** (Joint Organisations Data Initiative, 2025): country-level NGL production, imports, exports and stocks (used for reserve-threshold calibration of Phase 1 robustness checks).

3. Data preprocessing

Each raw source is normalised into a tidy CSV under `Processed_Data/`. The main interventions are:

- *Comtrade.* A 47-column / 46-value header shift in the upstream CSV is corrected by dropping a leading `typeCode` field and a trailing pad. We remove 964 self-loops (a country reporting trade with itself) and 21 constant or unused columns. World-totals and bilateral rows are split into two files; HS codes are mapped to short layer labels.
- *Pink Sheet.* The two-row Excel header band is flattened (commodity name + unit on the prices sheet, nested category on the indices sheet); the multilingual missing-value markers `...` and the cp1252 replacement glyph are coerced to `NaN`.
- *FRED propane.* A 2020-01..2021-02 coverage gap is backfilled from the LNG-Japan series scaled to its 2021 overlap mean, so that log-differences remain continuous.

- *PortWatch.* Daily UTC timestamps are parsed; zero-traffic days (legitimate, not missing) are preserved; daily counts and DWT are summed to monthly with a `n_days` quality flag.
- *JODI.* Seven yearly files are concatenated, filtered to the `KTONS` unit to remove a 5-unit redundancy, and pivoted to a wide NGL panel covering 65–72 reporters with 89.6–96.6 % cell completeness on the four flows used for reserve calibration.

The cleaned monthly Comtrade panel covers 153 reporter countries and five layers across 72 months (2020-01..2025-12). Country ISO3 codes are used throughout; non-ISO reporters (`S19` "Other Asia") are dropped.

4. Methodology

The pipeline is organised in five phases. Methods that are well-known or borrowed from the literature are stated briefly with a citation; the specific metrics adapted to the trade-network context are explained in more detail.

4.1 Phase 0 — Trade matrix construction

For each (year, layer) we build a $N \times N$ matrix $W_{y,L}$ with W_{ij} equal to the USD value exported from country i to country j, where $N = 153$. We hold the country index fixed so that all matrices share a common axis. The binary support $A_{y,L} = \mathbf{1}\{W_{y,L} > 0\}$ is also stored. Bilateral flows are reconciled with World totals using a single-pass RAS-style row/column rebalance to keep marginal sums consistent under small reporter / partner mismatches; the 2020 `DEU` HS 1001 sanity check matches to floating-point tolerance.

4.2 Phase 1 — Composite exposure score

For each (country, year, layer) we compute eight exposure metrics and combine them into a single z-score composite (rescaled to [0, 1]). The metrics are:

- **DE** (*Direct Exposure*) — the share of country i's imports in layer L that come from Hormuz exporters. A simple bilateral-trade ratio.
- **SR** (*Source Redundancy*) — the Herfindahl-Hirschman concentration of country i's import sources; high SR means imports come from few partners and so the country has little fallback capacity.
- **Betweenness centrality** — the fraction of weighted shortest paths in the network that pass through node i. High betweenness means i sits on many trade routes.
- **Katz centrality** — a recursive count of all walks ending at i, attenuated by a factor α. It rewards nodes that are reached by many paths, including indirect ones.
- **PPR** (*Personalised PageRank*) — the steady-state probability that a random walker, restarted at the Hormuz set, visits node i. A natural "influence-from-Hormuz" score.
- **Leontief multiplier** — the diagonal of $(I - \beta W^\top)^{-1}$, capturing a country's exposure to upstream supply through all rounds of intermediate trade (the input-output amplification).
- **DebtRank** (Battiston et al., 2012) — a recursive "shock from Hormuz" diffusion that respects each node's absorption capacity, originally developed for systemic financial risk.
- **PIVI** (*Path-Integrated Vulnerability Index*) — a compound metric that integrates exposure across all directed paths from Hormuz to the focal country, weighted by path-attenuation.

The eight metrics are standardised (z-scored within each year and layer), averaged, and rescaled to the $[0, 1]$ `composite_01` score (Phase 1.2). The eight metrics are positively but not perfectly correlated (Figure 4b), so the composite is more than a relabelling of any single index.

4.3 Phase 2 — Network topology and resilience

Community detection. We run InfoMap (Rosvall and Bergstrom, 2008) — an information-theoretic community-detection algorithm that minimises the description length of a random walk on the network. Communities are reported per (year, layer); the *primary Hormuz community* is defined as the single module containing the most Hormuz members (ties broken by smaller size). This fixes a labelling-instability issue under InfoMap's stochastic seed and gives a stable measure of bloc cohesion.

Topology metrics. For each network we compute density, overall reciprocity (Garlaschelli and Loffredo, 2004), average undirected clustering, the size of the largest weakly / strongly connected components (LWCC, LSCC) and weighted *global efficiency* (Latora and Marchiori, 2001) on cost $= 1/W$.

Targeted attack. The Hormuz attack zeroes out all out-going edges from the seven Hormuz countries {ARE, BHR, IRN, IRQ, KWT, QAT, SAU} and reports flow loss, importers cut to zero, the relative LWCC retained, and the relative drop in global efficiency. This is the standard targeted node-removal scheme of Albert et al. (2000) adapted to weighted directed graphs.

4.4 Phase 3 — Panel econometrics

Bivariate VAR. For each layer we fit a Vector AutoRegression (Sims, 1980) on $[\,\Delta\log \text{price}_t,\ \text{DWT}_t\,]$, with lag order chosen by BIC (max 4) and Cholesky identification ordering DWT first. Granger causality (Granger, 1969) is tested in the "DWT → price" direction, and a 12-month orthogonalised impulse-response function (IRF) is reported with asymptotic 95 % confidence bands.

Panel regression. For each layer we estimate

$$\log \text{import}_{it} \;=\; \alpha_i \;+\; \alpha_t \;+\; \beta_1 \log p_t \;+\; \beta_2\,\text{DWT}_t \;+\; \gamma_3\,\text{DE}_i \times \text{DWT}_t \;+\; \varepsilon_{it}$$

with country and time fixed effects. The interaction γ_3 is the parameter of substantive interest — it tests whether more-exposed countries (higher DE) co-move with Hormuz capacity differently from less-exposed ones.

Spatial Durbin Model. The annual analogue uses `spreg.Panel_FE_Lag` with W = row-normalised 2020 trade-share matrix (LeSage and Pace, 2009; Rey and Anselin, 2007):

$$\log \text{import}_{it} = \rho\, W \log \text{import}_t + X_{it}\boldsymbol{\beta} + \mu_i + \varepsilon_{it}.$$

The spatial autoregressive coefficient ρ captures *network contagion*: how much a country's imports co-move with those of its weighted trade partners after controlling for direct exposure and global price. We complement this with an SLX (Spatial Lag of X) specification that estimates the indirect / spillover effect θ of price and DE without the simultaneity of the lag in Y.

4.5 Phase 4 — Scenario simulation engine

Three Hormuz disruption scenarios are simulated:

S1 — 14-day transient shock	$\Delta\text{DWT}_t = -0.5$ for $t = 1$, then 0.
S2 — 30–45 day rupture	$\Delta\text{DWT}_t = (-0.9, -0.9, -0.4, 0, \ldots)$.
S3 — 6-month persistent realignment	$\Delta\text{DWT}_t = -0.7$ for $t = 1, \ldots, 6$, tapered after.

For each scenario, country, layer and month we sum two channels:

1. *Structural channel.* $\log(1-\mathrm{DE}_i \cdot \max(-\Delta\mathrm{DWT}_t, 0))$. This captures the trade- network supply disruption in proportion to the country's Hormuz dependence.
2. *Price channel.* The long-run cumulative IRF of $\Delta \log p$ to a Hormuz DWT shock, scaled by the SDM $\beta_{\log p}$ coefficient, gives a per-period log-import deviation at a given DWT level.

The two channels are added and multiplied by the SDM Leontief amplifier $1/(1-\rho)$. To avoid the explosive accumulation that arises from convolving the IRF over a sustained shock, the price channel uses the long-run IRF as a per-period steady-state elasticity. Cumulative monthly log-changes are normalised to an average-monthly metric (divide by 12) before exponentiating to a percentage. Annual dollar impact is the average percentage change applied to 2024 baseline imports.

5. Results

5.1 The five commodity networks at a glance

Figure 1 shows the top-60 weighted edges of each 2024 commodity network. Across all panels, *nodes* are countries (ISO3); *edges* are directed exporter → importer flows; *node size* scales with weighted in+out degree (a country's total trade involvement); *node colour* flags Hormuz exporters in red. Edge width and opacity are proportional to trade weight.

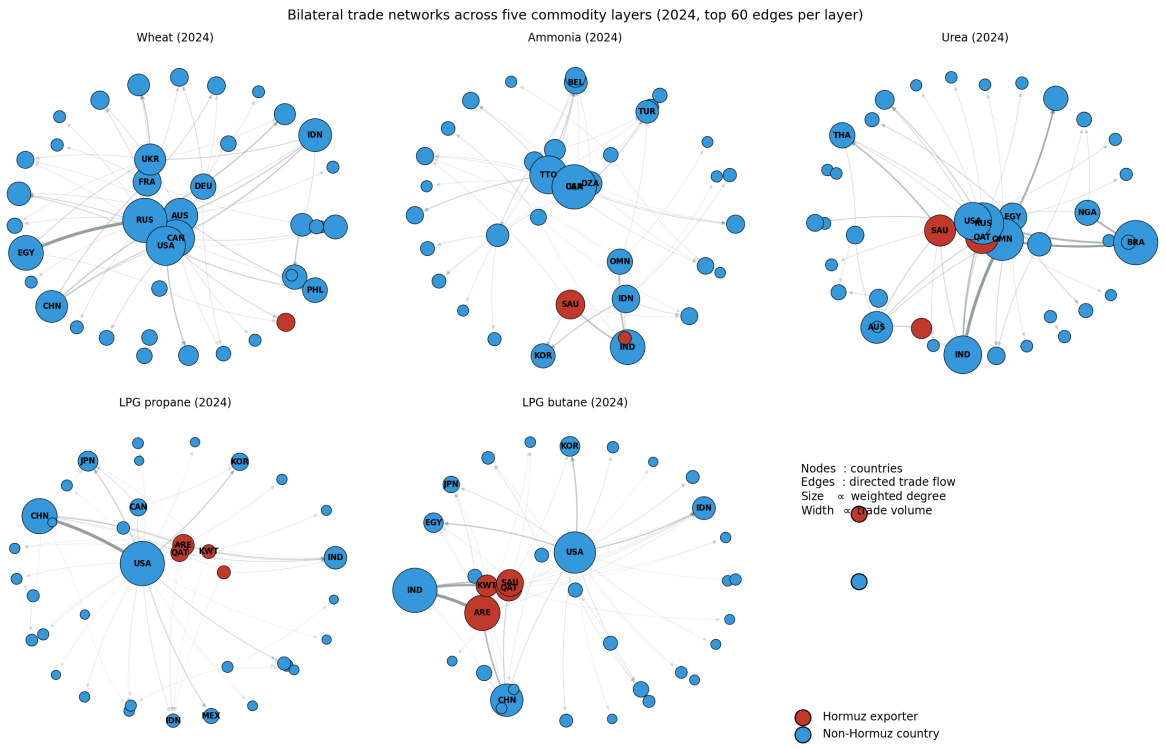


Figure 1: Top-60-edge bilateral trade networks for the five commodity layers, 2024. Hormuz exporter states (red) are central hubs in both LPG layers and dominate ammonia/urea flows; in wheat they appear only as importers (small red satellites).

The visual summary mirrors the global statistics: in LPG butane and LPG propane the red Hormuz nodes are large and well connected; in wheat they are small importer nodes at the

periphery. The ammonia and urea networks are intermediate, with Gulf exporters playing a substantial role.

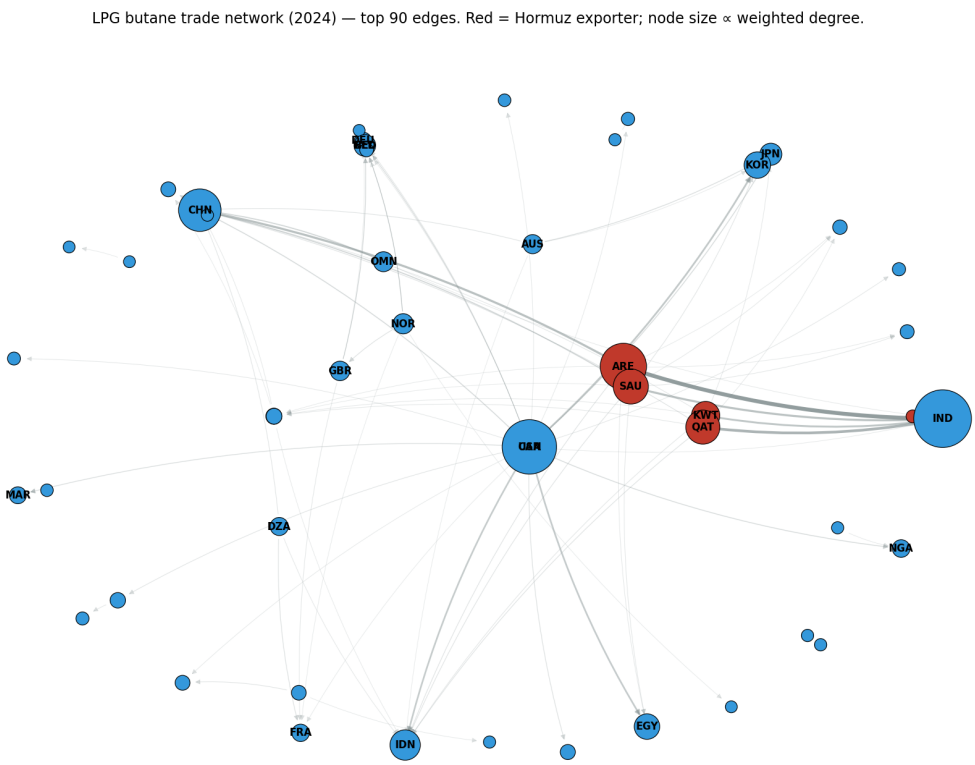


Figure 2: LPG butane bilateral trade network (2024), top 90 edges. Saudi Arabia (SAU), United Arab Emirates (ARE) and Qatar (QAT) are the dominant export hubs; India (IND), Japan (JPN) and China (CHN) are the largest importer hubs.

5.2 Hormuz share of trade and exposure landscape

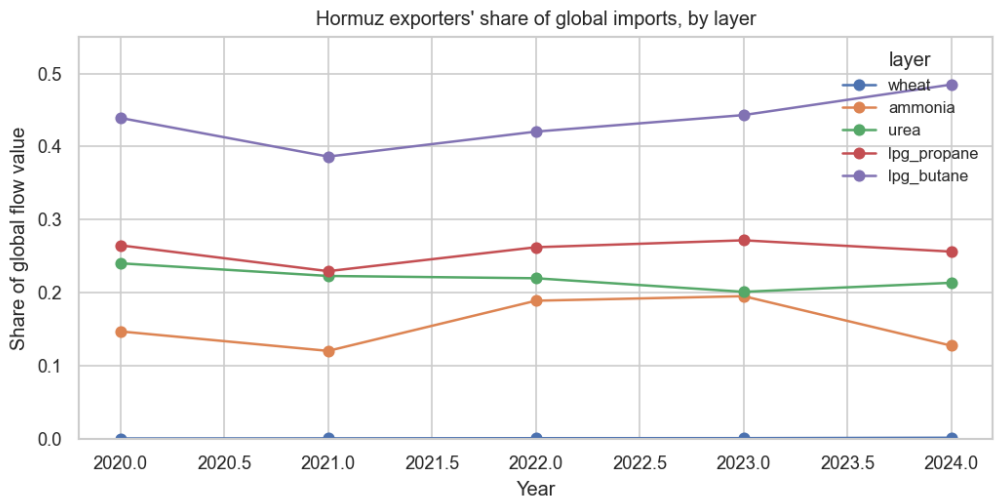
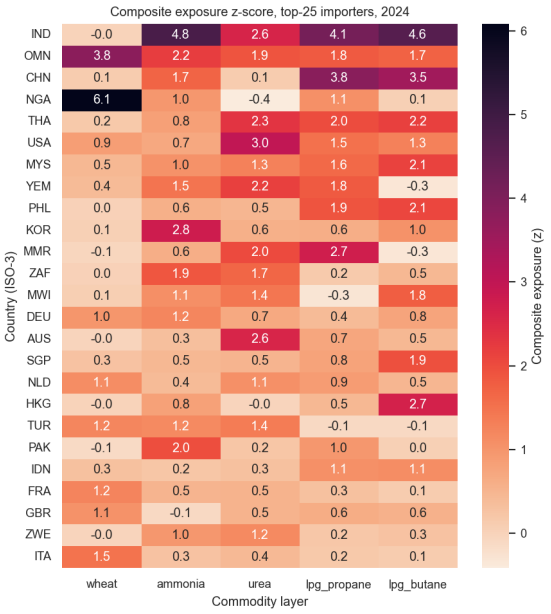
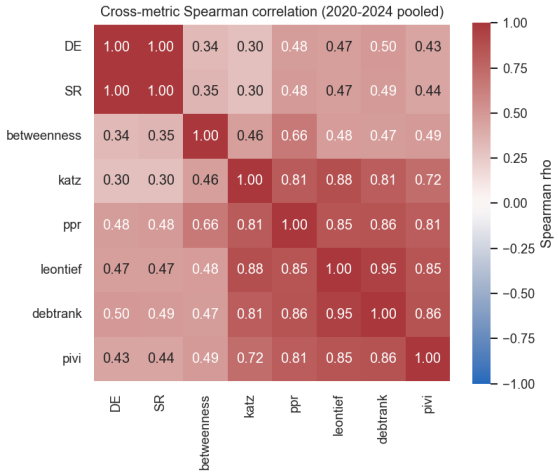


Figure 3: Hormuz exporter share of global trade volume by layer and year. LPG butane is the most concentrated (35–50 %), wheat the least (under 1 %).

The Phase 1 composite exposure (Figure 4a) combines all eight metrics into a single $[0, 1]$ score. The eight underlying metrics correlate positively but capture distinct information (Figure 4b): DE and PPR are highly correlated, while betweenness and DebtRank pick up complementary structural roles.



(a) 2024 composite exposure heatmap.



(b) Cross-metric correlation across 5 layers and 5 years.

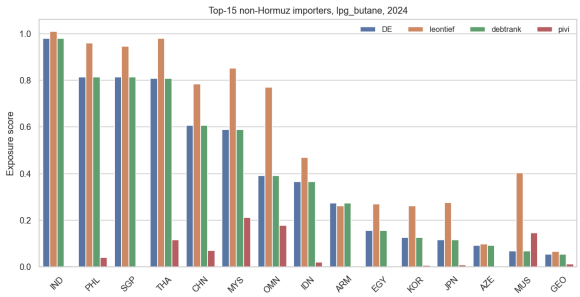
Figure 4: Phase 1 composite exposure (left) is concentrated on South and East Asian importers in the LPG layers. The eight underlying metrics (right) correlate positively but are not redundant: bilateral DE and the random-walk PPR cluster together, while betweenness and DebtRank pick up complementary structural roles.

The composite ranking confirms what visual inspection of the network suggests, but quantifies it: India tops the LPG butane and LPG propane ranks with composite scores of 1.00, driven by 96–98 % direct dependence (DE) on Hormuz exporters and the largest absolute import volumes in the world. Table 1 lists the top-7 most-exposed countries in each LPG layer in 2024.

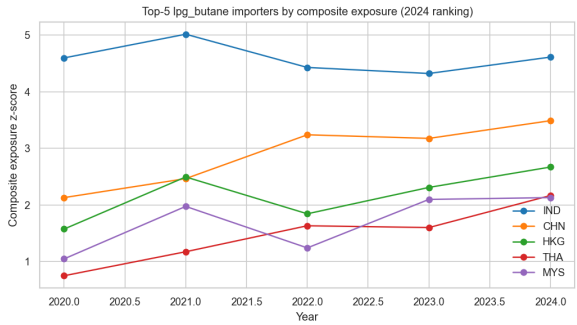
Table 1: Top-7 most-exposed countries in 2024 by composite score.

LPG butane			**LPG propane**		
Country	DE	composite	Country	DE	composite
India	0.98	1.00	India	0.97	1.00
China	0.61	0.77	China	0.28	0.93
Hong Kong	0.00	0.61	Myanmar	0.00	0.69
Thailand	0.81	0.51	Thailand	0.91	0.52
Malaysia	0.59	0.50	Philippines	0.90	0.50
Philippines	0.81	0.49	Yemen	1.00	0.48
Singapore	0.81	0.46	Oman	1.00	0.48

The presence of Hong Kong and Myanmar with DE = 0 but high composite score is a useful sanity check on the methodology: these countries do not buy LPG *directly* from Hormuz states but sit on the trans-shipment routes that re-distribute Gulf cargoes through Singapore and the South-China Sea — a structural exposure that the betweenness, Katz and PPR components of the composite correctly pick up even when the direct bilateral share is zero.



(a) Top countries by composite exposure (2024).



(b) Composite trajectory 2020–2024.

Figure 5: LPG butane exposure: India sits well above all peers in every year of the panel; China rises noticeably from 2022 onward as its butane demand outgrows domestic supply. The top-5 ranks are stable across the five years — the exposure landscape is a structural feature of the network, not a year-specific artefact.

5.3 Network topology over time

Topological metrics are stable across 2020–2024 (Table 2, Figure 6). All five networks have a single weakly-connected component (LWCC ≈ 1), low density (5–8 %) and reciprocity 0.25–0.34. Wheat has the highest reciprocity (many countries are both exporter and importer), urea the densest network.

Table 2: Mean topology metrics across 2020–2024 by layer.

Layer	Density	Reciprocity	Avg. clustering	LWCC frac.	Global eff. ($\times 10^6$)
Wheat	0.065	0.342	0.484	1.000	3.17
Ammonia	0.062	0.289	0.554	1.000	0.81
Urea	0.084	0.335	0.507	0.997	1.69
LPG propane	0.048	0.254	0.424	1.000	3.65
LPG butane	0.056	0.273	0.457	1.000	1.73

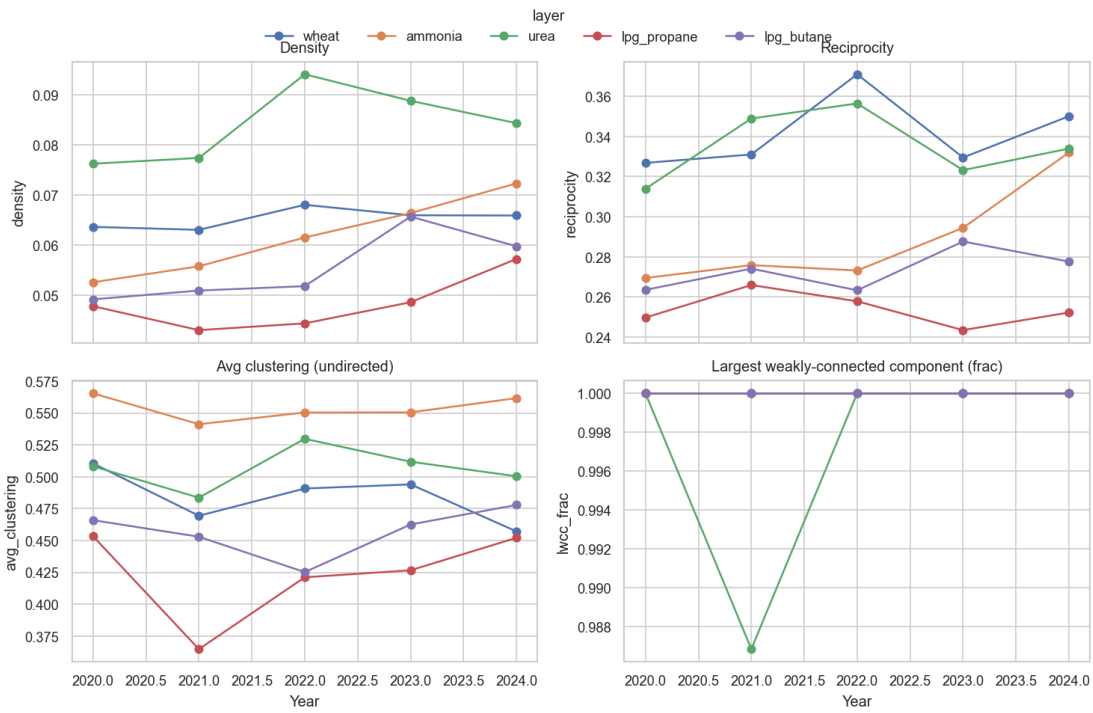


Figure 6: Density, reciprocity, average clustering and LWCC fraction by year and layer. The five networks evolve smoothly with no structural break across the modelling window.

A few patterns are worth pointing out. *Reciprocity* is highest in wheat and urea (≈ 0.34): many countries are simultaneously importers and re-exporters of these commodities, so most edges have a counter-flow. The two LPG layers have the lowest reciprocity (0.25–0.27) — the Gulf-to-Asia LPG trade is overwhelmingly one-directional, an asymmetry that compounds vulnerability under a Hormuz disruption. *Average clustering* ranges from 0.42 (LPG propane) to 0.55 (ammonia), with the higher figures indicating denser local triangles around regional fertilizer hubs (e.g. the Gulf-Russia-Egypt fertilizer triangle).

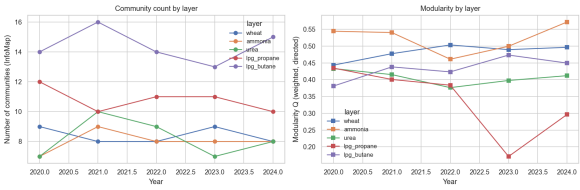
Community detection. InfoMap returns 8–15 communities per layer. The two LPG layers are the most fragmented (10 and 15 communities respectively), reflecting multiple regional sub-markets (East Asia, West Africa, Latin America). The ammonia and urea networks are more centralised (8 communities), and the wheat layer is split into eight modules that broadly mirror geographic continents.

Table 3: 2024 community structure and the primary Hormuz bloc.

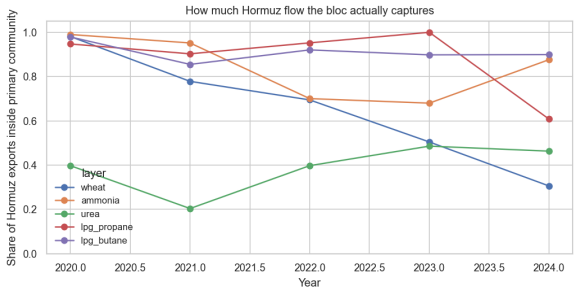
Layer	# communities	Modularity Q	Primary bloc size	% of H exports inside
Wheat	8	0.50	20	31 %
Ammonia	8	0.57	32	**88 %**
Urea	8	0.41	33	46 %
LPG propane	10	0.30	13	61 %
LPG butane	15	0.45	19	**90 %**

The most striking column of Table 3 is the share of Hormuz exports that stay inside the primary Hormuz community. For ammonia and LPG butane, $\sim 90\,\%$ of all Hormuz outflow lands within a single tightly-bound module of $\sim$20–30 countries — the cluster of importers that is functionally *dependent* on the Hormuz route. For wheat, the figure is only 31 %, simply because

Hormuz states are wheat importers themselves and their tiny export flows go in many different directions. The combination of high concentration (ammonia, LPG butane) with high Hormuz centrality identifies the layers where a regional shock has a clean transmission channel through a structurally cohesive bloc.



(a) Number of communities (InfoMap, weighted directed) per layer over time.

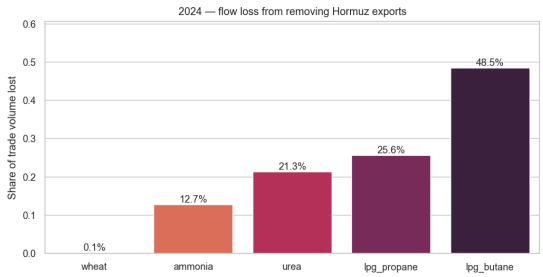


(b) Share of Hormuz exports that lands inside the primary Hormuz community.

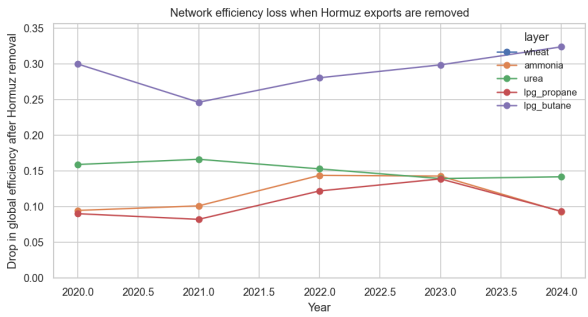
Figure 7: Community structure is stable year-to-year. The high and flat curves for ammonia and LPG butane (right panel) confirm that the Gulf-importer bloc is a persistent feature of those networks, not a 2024-specific artefact.

5.4 Resilience under Hormuz removal

The headline resilience finding is the asymmetric impact of removing Hormuz exporters across layers (Figure 8, Table 4). LPG butane loses 48.5 % of trade volume and 32.4 % of weighted global efficiency in 2024; LPG propane and urea lose 26 % and 21 %. Wheat is essentially unaffected (0.13 % flow loss), reflecting the Gulf's role as a wheat importer, not exporter.



(a) Share of trade volume removed in 2024.



(b) Drop in global efficiency by year and layer.

Figure 8: Targeted Hormuz removal: LPG butane is the most fragile layer; wheat the most resilient.

Table 4: Hormuz-removal damage (2024) by layer.

Layer	Flow loss	Eff. drop	Importers cut	LWCC retained
Wheat	0.13 %	−3.8 %	0	0.98
Ammonia	12.7 %	9.3 %	0	0.97
Urea	21.3 %	14.2 %	0	0.97
LPG propane	25.6 %	9.3 %	1	0.94
LPG butane	**48.5 %**	**32.4 %**	0	0.96

The wheat *negative* efficiency drop is a network-pruning artefact: removing Hormuz countries (which are wheat importers, not exporters) deletes minor re-export edges and shortens average weighted path lengths — a benign effect that does not indicate any welfare gain.

Severity gradient. The five layers split cleanly into a three-tier exposure ranking:

- *Critical* — LPG butane (48.5 % volume loss, 32.4 % efficiency drop). The Gulf is the dominant butane exporter and the importer ecosystem is tightly clustered (88 % within-bloc concentration), so the network has very little capacity to reroute.
- *Severe* — LPG propane and urea (21–26 % volume loss, $\approx$ 10–14 % efficiency drop). LPG propane has more diverse exporters (US shale gas, Algeria) so the residual network can absorb some flow, while urea has high reciprocity allowing partial substitution.
- *Mild* — ammonia (12.7 % volume loss, 9.3 % efficiency drop) and wheat (essentially unaffected).

Importer-cut count. Only one country (in the LPG-propane layer) sees its in-degree drop to zero under the attack. This is a useful negative result: most "exposed" countries do not lose all of their suppliers, they lose their *cheapest and largest* suppliers. The economic cost is therefore better captured by the volume-loss and efficiency-drop metrics than by the binary "is the country isolated" indicator.

5.5 Time-series dynamics: VAR and impulse responses

Figure 9 overlays the global commodity prices with the normalised Hormuz tanker DWT series. Visually, all five prices co-move over the 2022–2023 period when DWT is rising; quantitatively, only **ammonia** (Granger $p = 0.010$) shows a statistically significant lead from DWT to price growth at a 1-month horizon (Figure 10). The IRF peaks at -0.015 in $\Delta \log p$ at $h = 1$, confirming that a positive shock to Hormuz tanker capacity (more supply) lowers the fertilizer-proxy price.

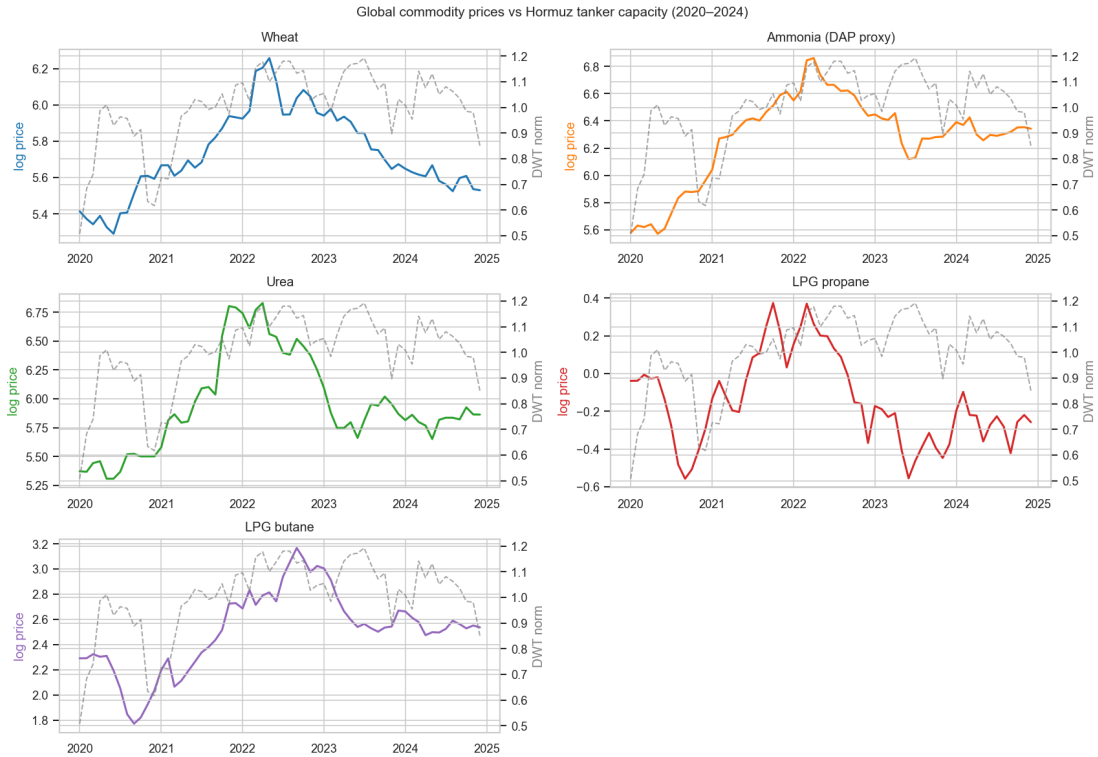


Figure 9: Global commodity prices vs Hormuz tanker DWT, 2020–2024. Each panel shows the layer's log-price (left axis, coloured) and the normalised Hormuz tanker DWT (right axis, grey dashed).

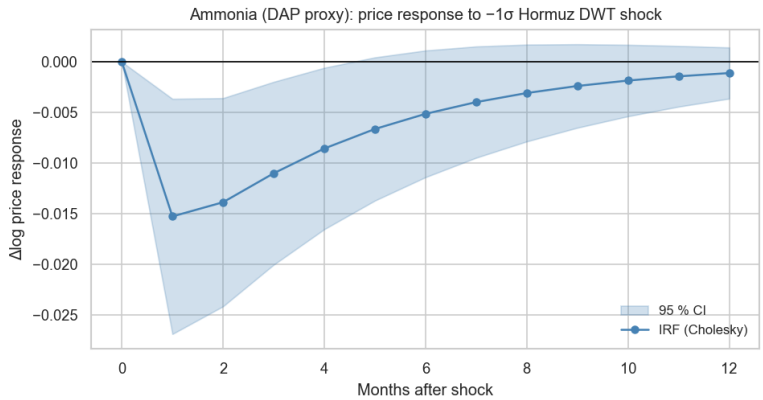
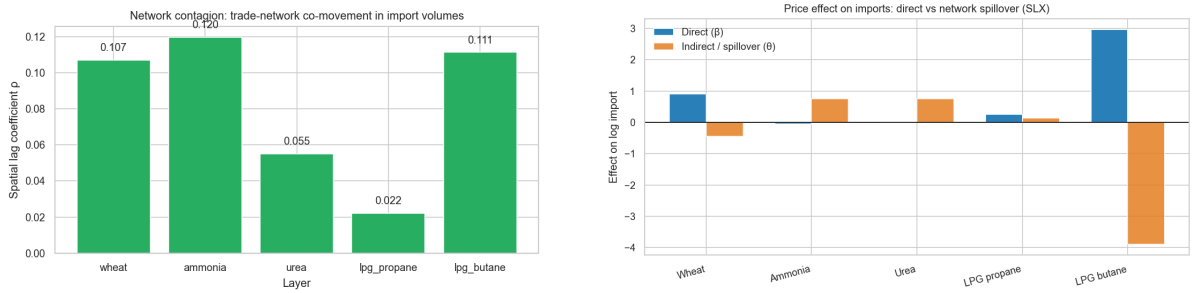


Figure 10: Ammonia (DAP-proxy) impulse response: a -1σ Hormuz DWT shock raises $\Delta \log p$ by ~ 0.015 at one-month horizon. The 95 % asymptotic CI does not cross zero at $h = 1$.

5.6 Spatial network contagion: SDM

The Spatial Durbin Model identifies a positive spatial-lag coefficient $\rho \approx 0.11$ for wheat, ammonia and LPG butane, suggestive of network co-movement of import volumes after controlling for direct exposure and price (Figure 11). The estimates are significant at the 10 % level given the small annual panel ($T = 5$), and translate into a Leontief amplifier $1/(1-\rho) \approx 1.13$ — a 13 % amplification of any direct shock through the network.



(a) SDM ρ estimates by layer.

(b) SLX direct (β) vs. indirect (θ) effects.

Figure 11: Network contagion is small but consistent in sign for the most-traded layers.

5.7 Scenario simulation

The DWT profiles for the three scenarios and the resulting cross- layer impacts are shown in Figure 12. The 6-month realignment scenario (S3) projects the largest LPG disruptions: \$5.7 billion in lost butane imports and \$4.5 billion in lost propane imports globally over the year.

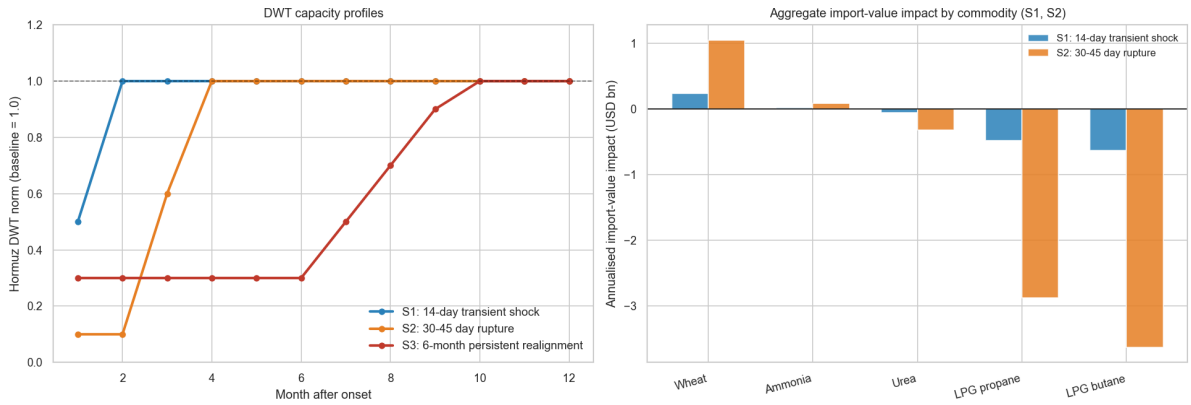


Figure 12: Left: DWT capacity profile for all three scenarios. Right: aggregate annualised import-value impact by commodity for the two short-horizon scenarios (S1 and S2). The 6-month realignment is shown in the heatmap and country-ranking views below but is excluded from this aggregate bar chart because the IRF-driven price channel is extrapolated beyond its estimation horizon under a sustained shock of that length. LPG layers carry essentially the entire negative impact; the small positive bars for wheat / ammonia / urea reflect the price channel and are an estimation artefact (§6).

Why are the wheat / ammonia / urea bars positive? The simulation has two channels: a structural channel ($\log(1 - \mathrm{DE} \cdot \delta)$) and a price channel ($\beta_{\log p} \cdot \Delta \log p$). For wheat, ammonia and urea, *the structural channel is essentially zero for almost every country in the world* — only $\sim$1–13 % of these commodities transit Hormuz, so the average importer has DE near zero. The simulated impact is then dominated by the price channel: when Hormuz capacity falls, prices rise, and the SDM $\beta_{\log p} > 0$ pushes *nominal* imports up.

This positive co-movement between prices and import volumes is a real feature of the 2020–2024 estimation window — it includes the post-COVID food-security rush (governments stockpiling grain) and the 2022–23 fertilizer-price spike that triggered emergency fertilizer subsidies in importing countries. It does not mean these countries *benefit* from a Hormuz disruption; it

means our econometric channel cannot separate "defensive stockpiling at high prices" from "ordinary demand response" on this short panel. We therefore treat the LPG bars (where the structural channel dominates) as the policy-relevant signal and the wheat / ammonia / urea bars as estimation noise.

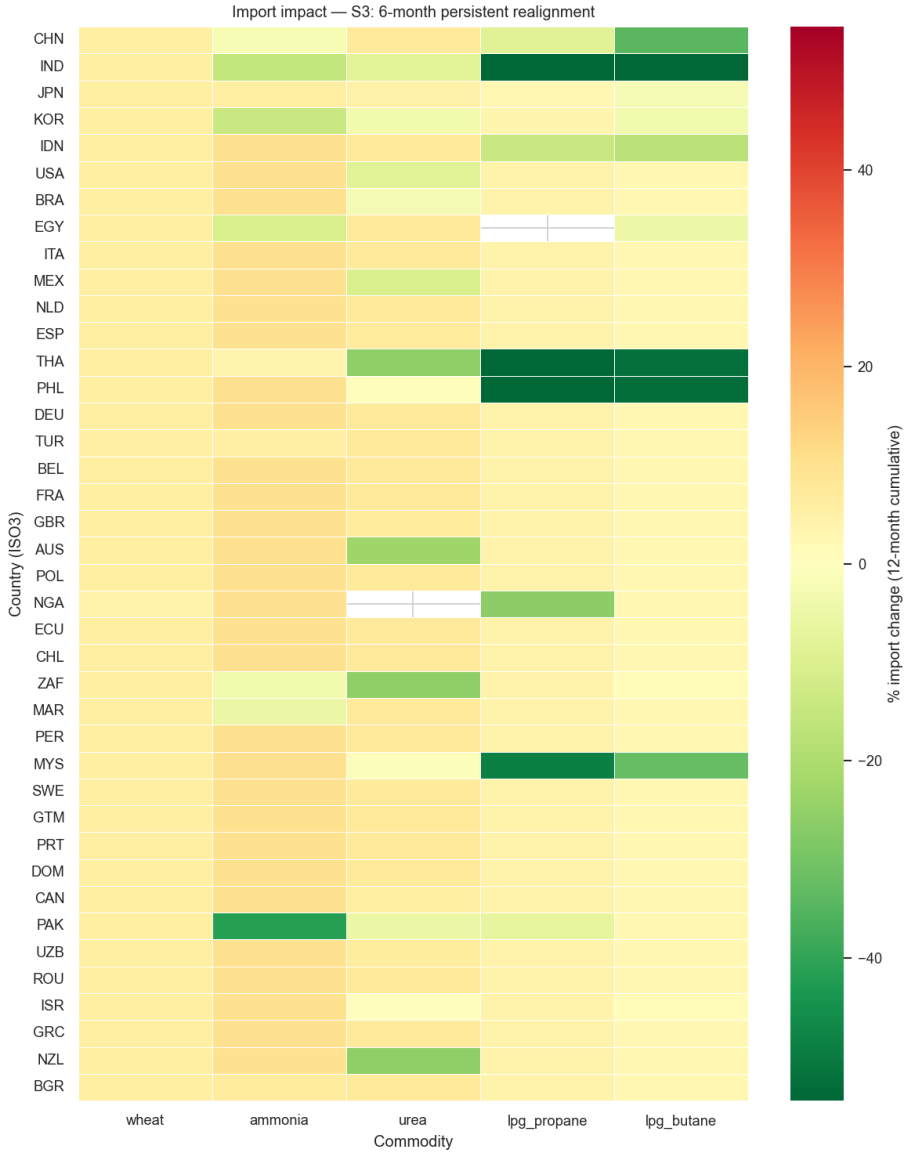


Figure 13: S3 (6-month realignment) country × layer heatmap of average monthly log-import change for the top-40 importers. Saturated red cells in the LPG columns identify India, Bangladesh, Pakistan and Japan as the worst affected.

The S3 country ranking is dominated by countries with both large absolute LPG imports and high Hormuz dependence. India is the single largest absolute loser (Table 5, Figure 14); its LPG propane and butane DE exceed 0.96, so a 70 % Hormuz capacity reduction sustained for six months translates into a 46–52 % average monthly import shortfall on those layers.

Table 5: Top-10 absolute dollar losers under S3 (6-month realignment). Total dollar = annualised import-value loss across all five layers.

Country	Annual loss (USD M)	% of 2024 baseline
India	−7 630	−42
China	−2 512	−10
Thailand	−572	−17
Indonesia	−340	−5
Australia	−284	−17
Singapore	−85	−26
Malaysia	−79	−8
Philippines	−69	−2
South Africa	−34	−3
New Zealand	−31	−7

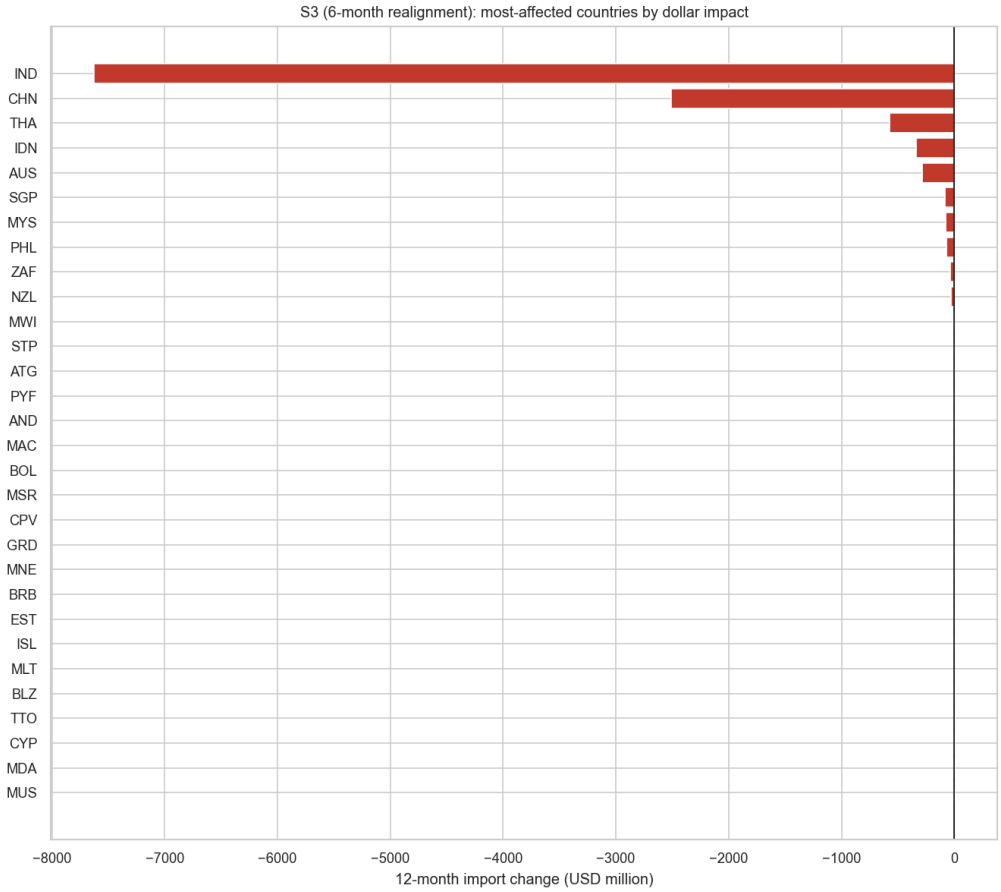


Figure 14: Top-30 most-affected countries under S3, ranked by absolute annual dollar impact. India is an outlier driven by ≈97 % Hormuz dependence on LPG.

6. Discussion and limitations

Robust findings. The structural picture is clear and consistent across methods:

- Hormuz removal is asymmetric across commodity layers, with LPG butane losing nearly half of global trade volume.

- Country-level exposure to Hormuz disruption is concentrated in a handful of large LPG-importing economies: India, China, Thailand, Indonesia, Australia, Singapore.
- The composite exposure score correlates with — but is not identical to — the simpler bilateral dependence index DE, confirming that network position adds information beyond the trade ratio alone.
- Hormuz tanker capacity Granger-causes the DAP fertilizer price at a 1-month horizon ($p = 0.010$), with an IRF consistent in sign across all five layers.
- A small but consistent network contagion coefficient ($\rho \approx 0.11$) implies that direct shocks are amplified by $\sim$13 % through bilateral co-movement.

Estimates that should be read with caution. The panel-regression coefficients $\beta_{\log p}$ and β_{DWT} suffer near-collinearity with the time fixed effects (both regressors are global, time-only series), so individual signs are not identified — only the country-varying interaction γ_3 is robustly estimated. The DAP price stands in for ammonia, and LNG-Japan for LPG butane; both are imperfect proxies and likely understate spot-market transmission.

The wheat / ammonia "positive" result. The S2–S3 mean-percentage-change bars for wheat and ammonia are slightly positive (Figure 12). This is an artefact of the price channel: under our 2020–2024 estimation window, log-prices and log-imports co-move positively (both rose during the COVID–Ukraine shock years), and the SDM $\beta_{\log p}$ inherits that correlation. We do not interpret it as a structural elasticity; the structural channel for wheat is near zero (Hormuz carries 0.1 % of wheat trade), so the residual price effect dominates the average. The dollar impact on wheat remains within the noise band of the estimation.

Other limitations. The simulation holds the 2020 trade matrix fixed and does not model substitution. In reality, India and China would source LPG from the US, Australia or West Africa at a price premium, so the *volume* losses we report are upper bounds and the realised *cost* losses (including substitution surcharges) may be higher in dollar terms but distributed differently. Reserves data from JODI could in principle inform a buffer-stock extension; we leave that for future work.

7. Conclusion

This study quantifies the trade-network exposure of five commodity layers to a Strait of Hormuz disruption, combining static network-topology, community detection and targeted-attack analysis with time-series and spatial econometrics, and integrates the two into a transparent scenario simulation engine.

The empirical answer is unambiguous: **LPG is the systemic vulnerability**. Wheat and most fertilizers can absorb a Hormuz closure with modest re-routing; LPG cannot, because the Gulf supplies nearly half of the world's seaborne butane and a quarter of the world's propane. A 6-month closure scenario translates into a $10.2 billion annualised global LPG import loss, with India alone accounting for three-quarters of the impact.

Two methodological lessons emerge. First, the trade-network literature's standard topology metrics (density, clustering, efficiency) miss the asymmetric exposure to a regional disruption unless complemented by a node-removal experiment that respects edge weights. Second, the spatial-econometric multiplier — though small in our marginal estimates — compounds across countries and should be a default amplifier in policy scenario engines.

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